

SCI RESEARCH ON THE TRANSLATIONAL SPECTRUM LETTER OF INTENT



Respond to each numbered item using the boxes provided. Complete responses using Arial 11-point font, single-spaced, with double spacing between paragraphs. Do not alter margins, delete form text, or exceed the page limits provided in this form.

1	PI/Applicant Name	Eric Meyers		
2	Organization	The University of Texas at Dallas		
3	Mentor Name(s) (for Postdoctoral Fellowships)	N/A		
4	Project Title	Targeted neuromodulation therapy to improve walking after spinal cord injury		
5	Funding Category	Translational Spectrum Grants	☐	2-year grant term
			X	3-year grant term (human subjects required)
		Postdoctoral Fellowships	☐	2-year grant term
6	Total Amount Requested (Round to nearest whole dollar)	\$400,000		
7	Budget Summary Provide a brief outline of the proposed cost categories (e.g., personnel, supplies, travel, animals, other). Limit the response to 4-5 lines of text. This budget may be modified at the full grant application (FGA) stage.	The requested budget supports personnel costs including the PI (10% effort), a clinical trial manager for IDE amendment support and clinical trial management, physical therapist, a biostatistician, 1 full time graduate student, the purchase of consumables for the EMG and stimulation systems, participant stipends for sessions, and dissemination expenses for open-access publication and data deposition.		
8	Research Subject(s) Select the option that best describes your research subject(s).	☐	Animals	
		X	Humans	
		☐	Animals and Humans	
		☐	Other (please specify):	
9	Is this a resubmission of a previously reviewed (but not funded) Neilsen Foundation Full Grant Application (FGA)?	☐	Yes	X No
Enter all responses inside the appropriate boxes. Following the instructions above, respond to sections 10–12 using only pages two and three of your completed form. Items 13 & 14 (if applicable) should begin on page four and adhere to page limitations as described. A bibliography is <u>not</u> required for the Letter of Intent (LOI) but one may be attached as a single page at the end of this form and does not count toward the three-page limit.				

10	State your hypotheses and project Aim(s). Hyperactive stretch-reflex circuitry, driven by the loss of descending inhibition and maladaptive spinal plasticity, is a principal biomechanical bottleneck that prevents many individuals with motor-incomplete spinal cord injury (SCI) from regaining an efficient, pain-free gait¹⁻³. Pharmacologic options such as baclofen, tizanidine, and intramuscular botulinum toxin dampen reflex excitability but also weaken residual voluntary drive, limiting functional gains⁴. Our collaborative team at the National Center for Adaptive Neurotechnologies (NCAN) has developed a targeted plasticity approach that involves operant down-conditioning of the soleus H-reflex as a learning-based alternative treatment⁵⁻⁷. By progressively rewarding trials in which the reflex is smaller than a criterion, patients reduce reflex excitability, reduce ankle stiffness, and walk faster—even years after injury⁷. However, the reliance of H-reflex operant conditioning (HROC) on explicit performance-based rewards using real-time visual feedback demands continuous attention. This can be cognitively demanding for individuals particularly for the therapy durations required (1-hour sessions, 3x week for 12 weeks), thus limiting compliance and limiting participation to cognitively intact individuals. Importantly, the therapy is currently tethered to research laboratories thus limiting home-based therapy. Our team at the University of Texas at Dallas (UTD) has pioneered the use of pairing vagus nerve stimulation (VNS) with rehabilitation to enhance plasticity and recovery after various forms of neurological injury, including SCI. VNS provides precisely timed engagement of neuromodulatory systems^{8,9} that in turn drive rapid, phasic release of pro-plasticity molecules which are known to govern synaptic plasticity throughout brain and spinal networks¹⁰. Pairing the VNS-dependent release of neuromodulators with the repeated activation of CNS networks drives robust plasticity in these networks^{9,11,12}. Importantly, our recent data shows that VNS can facilitate synaptic plasticity with minimal conscious awareness^{13,14}. We therefore posit that substituting HROC's visual rewards with a VNS-based teaching signal could convert H-reflex conditioning from a cognitively taxing, clinic-bound protocol into a fully automated intervention suitable for home use. Embedding the reinforcement signal in an implanted device retains the mechanistic specificity of operant conditioning while eliminating the need for prolonged concentration. Aim 1 will establish the safety and feasibility of VNS-reinforced HROC (V-HROC) in six adults with chronic (≥ 12 mo) AIS C/D spinal cord injury and gait impairments. All candidates are drawn from our existing pool of participants who were previously implanted with a VNS system during an early feasibility study¹⁵. Therefore, no additional surgery is required, greatly reducing risk and start-up time. Success requires at least 80% session completion and no device-related serious adverse events during the 12-week intervention. Aim 2 will test early efficacy by comparing pre- versus post-therapy soleus H-reflex amplitude, walking speed, WISCI-II, and Modified Ashworth scores. We anticipate significant reflex depression, faster gait, and reduced spasticity persisting four weeks after treatment, thereby providing preliminary evidence of VNS as a practical reward substitute. Aim 3 will elucidate the mechanisms of recovery by using motion-capture analysis to quantify pre- versus post-therapy changes in gait kinematics and overall movement quality.
11	Provide an overview of the experimental design/methods. The proposed study is a prospective, single-arm pilot trial that will recruit six adults (18-65 yr) with chronic (≥ 12 mo) motor-incomplete spinal cord injury (AIS C/D) who present with clinically significant gait impairments. Initially, we will secure regulatory approval to conduct the pilot study by amending our existing IRB and IDE protocols to incorporate the combination of HROC with VNS. Given the low-risk profile of HROC, we anticipate an expedited approval process, and our team has experience amending our active IDE for VNS in SCI to incorporate other therapy modalities. After informed consent, each participant completes two baseline visits during which we capture reference measures of reflex excitability (soleus H- and M-wave recruitment curves), spasticity (Modified Ashworth Scale), and functional mobility (10-m and 6-min Walk Tests, Walking Index for Spinal Cord Injury-II, and marker-based motion-capture gait metrics). The intervention spans twelve weeks and consists of three conditioning sessions per week, each lasting approximately one hour. Throughout the trial the participant stands while surface EMG from the soleus muscle is processed in real time. When background EMG is held within a low target window for

	two seconds, the tibial nerve is stimulated to evoke an H-reflex. On trials in which the reflex amplitude falls below a session-specific criterion (chosen so that ~50 % of trials meet the target), the implant immediately delivers a 0.5-s train of VNS (0.8 mA, 30 Hz, 100 μs pulse width; parameters known to facilitate neuroplasticity). No visual or auditory feedback of performance is provided. The burst of VNS serves as the only reinforcement signal, thereby testing the central hypothesis that neuromodulator release can replace conscious reward cues and enable attention-free, potentially home-based conditioning. Each visit comprises 240 reflex trials, yielding roughly 120 contingent VNS deliveries—a dosing paradigm matched to our prior work in both HROC and VNS. To control for non-associative effects, the first two weeks employ a “control” version of the protocol in which reflexes are recorded but no reinforcement (VNS) is delivered; these data provide a within-subject baseline. Functional outcomes are repeated at week 6 (mid-therapy), week 12 (end of therapy), and four weeks after therapy to examine durability. Safety is monitored continuously; adverse events are reviewed by an independent physician. Feasibility is defined as completion of the target 120 VNS deliveries in at least 80% of scheduled sessions. Primary efficacy endpoints are the percent change in soleus H-reflex amplitude and walking speed. Secondary endpoints include WISCI-II score, Modified Ashworth grade, cadence, stride length, and step-length asymmetry. Exploratory motion-capture analyses will examine stance-swing ratio, double-support time, step-width variability, and key joint angles (hip extension, swing-phase knee flexion, push-off ankle plantar-flexion) to clarify balance and propulsion changes. Outcomes will be analyzed with linear mixed-effects models that treat time as a fixed factor and participant as a random factor. Video recordings of all clinical assessments will be scored by blinded raters to minimize bias. Data analysis will be performed in Python or MATLAB, and de-identified datasets will be released on OpenNeuro six months after study completion, ensuring transparency and reproducibility. This pragmatic design rigorously evaluates whether contingent VNS can act as an intrinsic reward signal to drive reflex down-conditioning and improve gait while laying the groundwork for a future, randomized controlled trial.
12	Explain the project’s relevance to the Neilsen Foundation. The proposed study addresses one of the largest barriers to mobility after incomplete SCI—hyperactive stretch reflexes that generate inefficient gait—with a therapy that is mechanistically grounded and immediately translatable. By substituting a precisely timed burst of VNS for the visual reward normally used in HROC, the project leverages an existing FDA-approved neuromodulatory therapy¹⁶ to create an intrinsic, attention-free reinforcement signal. Eliminating the need for conscious engagement removes a major obstacle that has so far limited HROC to highly supervised laboratory settings. It opens the door to home-based, scalable rehabilitation that can reach rural and underserved populations the Neilsen Foundation is committed to supporting. The work sits well within the SCIRTS Pilot Clinical category. It builds on decades of compelling pre-clinical and randomized controlled clinical trials, rigorously evaluates safety and feasibility in a well-defined cohort, and generates the preliminary efficacy evidence required for a subsequent randomized trial. Because all participants are already implanted with the ReStore VNS device from our previous study¹⁵, and our team will amend our existing IDE, the project is cost-efficient and low-risk, allowing Neilsen resources to be leveraged for high-impact questions rather than surgical and regulatory costs. The protocol’s reliance on objective electrophysiological and functional outcomes, its preregistered open-data plan, and its close integration with a multidisciplinary rehabilitation team at UT Dallas and Baylor Research Institute embody the Foundation’s emphasis on rigor, transparency, and clinical relevance. If contingent VNS can indeed replace external rewards while amplifying therapeutic benefit, the findings will redefine how HROC is delivered. It will provide the SCI community with a practical, non-pharmacological option to regain safer, faster walking and greater autonomy. It will also justify exploring the use of VNS-reinforced HROC (V-HROC) in the upper limb. These goals align precisely with the Neilsen Foundation’s mission to change lives after spinal cord injury.
13	Conflict of Interest (COI) Disclosure—Clearly and briefly describe all perceived or real COIs. <i>(Excluded from the three-page limit. Enter N/A if you have nothing to disclose.)</i> N/A

14	Resubmission: If this LOI is the <u>first</u> resubmission of a previously reviewed but not funded <u>FGA</u> , up to a half-page response to reviewer comments may be entered here. (<i>Excluded from the three-page limit.</i>) N/A
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